

BANK OF COBOL

Core Banking System — Business Rules

Migration Reference Document

Source Program:	BANKING-SYSTEM (BANKING.cbl)
Platform:	IBM Mainframe (COBOL)
Prepared for:	System Migration — Business Analyst Review
Operations:	Deposit and Withdrawal only

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1. Account Data Model

Each client account holds the following information:

Field	Description
Client ID	Unique 6-digit number assigned at account opening (starts at 100001)
Last Name	Up to 15 characters
First Name	Up to 10 characters
Balance	Current account balance

Field	Description
Status	ACTIVE, FROZEN, or CLOSED
Account Type	REGULAR or VIP
Daily Withdrawal Total	Cumulative withdrawals processed today (resets at end of day)
Daily Deposit Total	Cumulative deposits processed today (resets at end of day)
Date Opened	The date the account was opened
Total Transaction Count	Lifetime count of all transactions (approved and rejected)
Last Transaction Date	Date of the most recent transaction attempt
Last Transaction Type	Whether the last transaction was a Deposit or Withdrawal
Last Transaction Amount	Dollar amount of the most recent transaction

2. Account Opening & Initialization

- BR-001 — Client ID Assignment**

Every new account is assigned a unique 6-digit Client ID. IDs are assigned sequentially, beginning at 100001.
- BR-002 — Opening Balance**

All new accounts are funded with an initial balance of \$500.00 at the time of opening.
- BR-003 — Account Type at Opening**

At the time of opening, the system evaluates the account balance to determine account type: if the opening balance exceeds \$50,000.00, the account is classified as VIP; otherwise Regular.

Note: With a standard opening balance of \$500.00, all new accounts open as Regular.
- BR-004 — Daily Accumulators at Opening**

Daily withdrawal and deposit totals are initialized to \$0.00 at account opening.

3. Account Status Management

An account can be in one of three statuses. Status governs whether transactions are permitted.

Status	Code	Description
Active	A	Account is in good standing. All permitted transactions can be processed.
Frozen	F	Account has been suspended. No deposits or withdrawals are allowed. Customer must visit a branch to unfreeze.
Closed	C	Account is permanently closed. No transactions of any kind are permitted.

BR-005 — Frozen Account Restriction

No transaction (deposit or withdrawal) may be processed on a Frozen account.

"ACCOUNT FROZEN - VISIT BRANCH TO UNFREEZE"

BR-006 — Closed Account Restriction

No transaction may be processed on a Closed account.

"ACCOUNT IS PERMANENTLY CLOSED"

BR-007 — Status Check is First Priority

Account status is always validated before any amount or limit checks. If an account is Frozen or Closed, the transaction is immediately rejected with no further validation.

4. Account Tier Classification (Regular vs VIP)

BR-008 — VIP Qualification Threshold

An account qualifies for VIP status when the account balance exceeds \$50,000.00.

BR-009 — Regular Account

Any account with a balance of \$50,000.00 or less is classified as Regular and is subject to Regular transaction limits.

BR-010 — VIP Account

Any account with a balance above \$50,000.00 is classified as VIP and is subject to VIP transaction limits (significantly higher).

5. Deposit Processing

Deposits are validated in the following sequence. The first failure immediately rejects the transaction.

5.1 Deposit Validation Sequence

Step 1 — Account Status Check

- If account is FROZEN → Reject (see BR-005)
- If account is CLOSED → Reject (see BR-006)

Step 2 — Minimum Amount Check**BR-011 — Minimum Deposit Amount**

The minimum deposit amount is \$1.00.

"AMOUNT BELOW MINIMUM DEPOSIT OF \$1.00"

Step 3 — Single Transaction Cap**BR-012 — Regular Account: Single Deposit Limit**

A Regular account may not receive more than \$50,000.00 in a single deposit.

"EXCEEDS REGULAR SINGLE DEPOSIT CAP \$50,000"

BR-013 — VIP Account: Single Deposit Limit

A VIP account may not receive more than \$250,000.00 in a single deposit.

"EXCEEDS VIP SINGLE DEPOSIT CAP \$250,000.00"

Step 4 — Daily Cumulative Cap**BR-014 — Regular Account: Daily Deposit Limit**

The total of all deposits to a Regular account in a single business day may not exceed \$100,000.00.

"DAILY DEPOSIT LIMIT REACHED \$100,000.00"

BR-015 — VIP Account: Daily Deposit Limit

The total of all deposits to a VIP account in a single business day may not exceed \$500,000.00.

"VIP DAILY DEPOSIT LIMIT REACHED \$500,000"

5.2 Approved Deposit Processing

When a deposit passes all validations, the system:

- Adds the deposit amount to the account balance.
- Adds the deposit amount to the account's daily deposit accumulator.
- Increments the transaction counter for the account by 1.
- Records the last transaction date, type (Deposit), and amount on the account.
- Evaluates whether the new balance qualifies the account for VIP upgrade (see BR-016).

6. Withdrawal Processing

Withdrawals are validated in the following sequence. The first failure immediately rejects the transaction.

6.1 Withdrawal Validation Sequence

Step 1 — Account Status Check

- If account is FROZEN → Reject (see BR-005)
- If account is CLOSED → Reject (see BR-006)

Step 2 — Minimum Amount Check**BR-020 — Minimum Withdrawal Amount**

The minimum withdrawal amount is \$1.00.

"AMOUNT BELOW MINIMUM WITHDRAWAL OF \$1.00"

Step 3 — Single Transaction Cap**BR-021 — Regular Account: Single Withdrawal Limit**

A Regular account may not withdraw more than \$10,000.00 in a single transaction.

"EXCEEDS REGULAR SINGLE WITHDRAWAL CAP \$10K"

BR-022 — VIP Account: Single Withdrawal Limit
A VIP account may not withdraw more than \$50,000.00 in a single transaction.
"EXCEEDS VIP SINGLE WITHDRAWAL CAP \$50,000"

Step 4 — Daily Cumulative Cap

BR-023 — Regular Account: Daily Withdrawal Limit
The total of all withdrawals from a Regular account in a single business day may not exceed \$20,000.00.
"DAILY WITHDRAWAL LIMIT REACHED \$20,000.00"

BR-024 — VIP Account: Daily Withdrawal Limit
The total of all withdrawals from a VIP account in a single business day may not exceed \$100,000.00.
"VIP DAILY WITHDRAWAL LIMIT REACHED \$100,000"

Step 5 — Balance Adequacy Checks

BR-025 — Minimum Required Balance
After a withdrawal, the account balance must not fall below \$100.00.
"WOULD BREACH MINIMUM REQUIRED BALANCE \$100.00"

BR-026 — No Overdraft
The system does not permit overdrafts. A negative post-withdrawal balance is not allowed.
"INSUFFICIENT FUNDS - OVERDRAFT NOT PERMITTED"
Note: BR-025 is checked first. BR-026 is only reached if the minimum balance were \$0.00.

6.2 Approved Withdrawal Processing

When a withdrawal passes all validations, the system:

- Deducts the withdrawal amount from the account balance.
- Adds the withdrawal amount to the account's daily withdrawal accumulator.
- Increments the transaction counter for the account by 1.
- Records the last transaction date, type (Withdrawal), and amount on the account.
- Evaluates whether the new balance requires VIP → Regular downgrade (see BR-017).

7. Transaction Limits Reference Table

Limit	Regular Account	VIP Account
Minimum transaction amount	\$1.00	\$1.00
Maximum single deposit	\$50,000.00	\$250,000.00
Maximum single withdrawal	\$10,000.00	\$50,000.00
Maximum daily deposits (total)	\$100,000.00	\$500,000.00
Maximum daily withdrawals (total)	\$20,000.00	\$100,000.00
Minimum required balance	\$100.00	\$100.00

Limit	Regular Account	VIP Account
Overdraft permitted	No	No

8. Account Tier Promotion & Demotion

BR-016 — Automatic VIP Upgrade After Deposit

After a deposit is applied, if the new balance exceeds \$50,000.00 and the account is currently Regular, the account is automatically upgraded to VIP. New VIP limits apply immediately.

BR-017 — Automatic VIP Downgrade After Withdrawal

After a withdrawal is applied, if the new balance is \$50,000.00 or less and the account is currently VIP, the account is automatically downgraded to Regular. The operator is notified: "NOTE: ACCOUNT DOWNGRADED VIP -> REGULAR".

BR-018 — Tier Applies at Time of Transaction

The account type at the moment of transaction submission determines which limits apply. A mid-day upgrade or downgrade takes effect immediately for all subsequent transactions.

9. Transaction Logging & Audit Trail

BR-030 — All Transactions Are Logged

Every transaction attempt — whether approved or rejected — is recorded in the transaction log. A rejected transaction is not applied to the balance but is still written to the log.

BR-031 — Transaction Log Contents. Each log entry captures:

- Transaction ID (sequential, system-generated)
- Transaction Date and Time
- Client ID
- Transaction Type (Deposit or Withdrawal)
- Amount
- Account Balance after the transaction (approved transactions)
- Result: Approved (OK) or Rejected (RJ)
- Rejection Reason (if rejected)

BR-032 — Circular Log Buffer

The transaction log holds a maximum of 5,000 entries. When full, new entries overwrite the oldest entries (circular buffer). The total count is retained across the session.

BR-033 — Transaction IDs are Sequential

Each log entry is assigned a unique, sequentially incrementing Transaction ID starting at 1. IDs are never reused within a session.

10. End-of-Day Processing

BR-040 — Daily Limit Reset

At the end of each business day, an authorized operator must manually trigger the end-of-day reset. This clears daily withdrawal and deposit accumulators to \$0.00 for every account and marks the beginning of a new business day.

BR-041 — Confirmation Required

The end-of-day reset requires explicit operator confirmation (Y/N prompt) before executing. If the operator does not confirm, the reset is cancelled and no data is changed.

BR-042 — Reset Applies to All Accounts

The daily reset applies simultaneously to all accounts regardless of status or type. Frozen and Closed accounts also have their daily accumulators cleared.

BR-043 — Daily Limits are Per Business Day

Transaction limits are calculated from the start of the current business day. After an end-of-day reset, every account starts the new day with \$0.00 accumulated toward daily limits.

11. Inquiries & Reporting

11.1 Balance Inquiry (BR-050)

Any account (including Frozen and Closed) may be queried for its current balance. Displays:

- Current balance
- Total amount withdrawn today
- Total amount deposited today
- Lifetime transaction count

11.2 Account Detail (BR-051)

A full account record view displays all account fields, plus the applicable transaction limits based on current account type (Regular or VIP), and the minimum required balance.

11.3 Client List (BR-052)

The client list is displayed 20 records at a time. The operator selects a starting record number (1 to 1000). Shows Client ID, name, balance, status, and account type for each account.

11.4 Daily Summary Report (BR-053)

The daily summary provides an aggregate view for the current business day:

- Total deposits across all accounts today
- Total withdrawals across all accounts today
- Total number of transactions logged in the session
- Count of accounts by status: Active, Frozen, Closed

11.5 Transaction Log View (BR-054)

The operator can view the most recent 20 transactions. Entries show: Transaction ID, date, Client ID, transaction type, amount, and result (OK/RJ).

12. System Capacity & Constraints

Parameter	Value
Maximum accounts in database	1,000
Maximum transaction log entries	5,000 (circular — oldest overwritten)
First assigned Client ID	100001
Supported transaction types	Deposit and Withdrawal only
Overdraft facility	Not supported
Inter-account transfers	Not supported
Interest calculation	Not supported
Account creation via system	Not supported (pre-loaded database)
Account closure via system	Not supported (status is pre-set)

13. Migration Notes for Business Analysts

The following values are hardcoded in the COBOL source and must be externalized as configurable parameters in the new system:

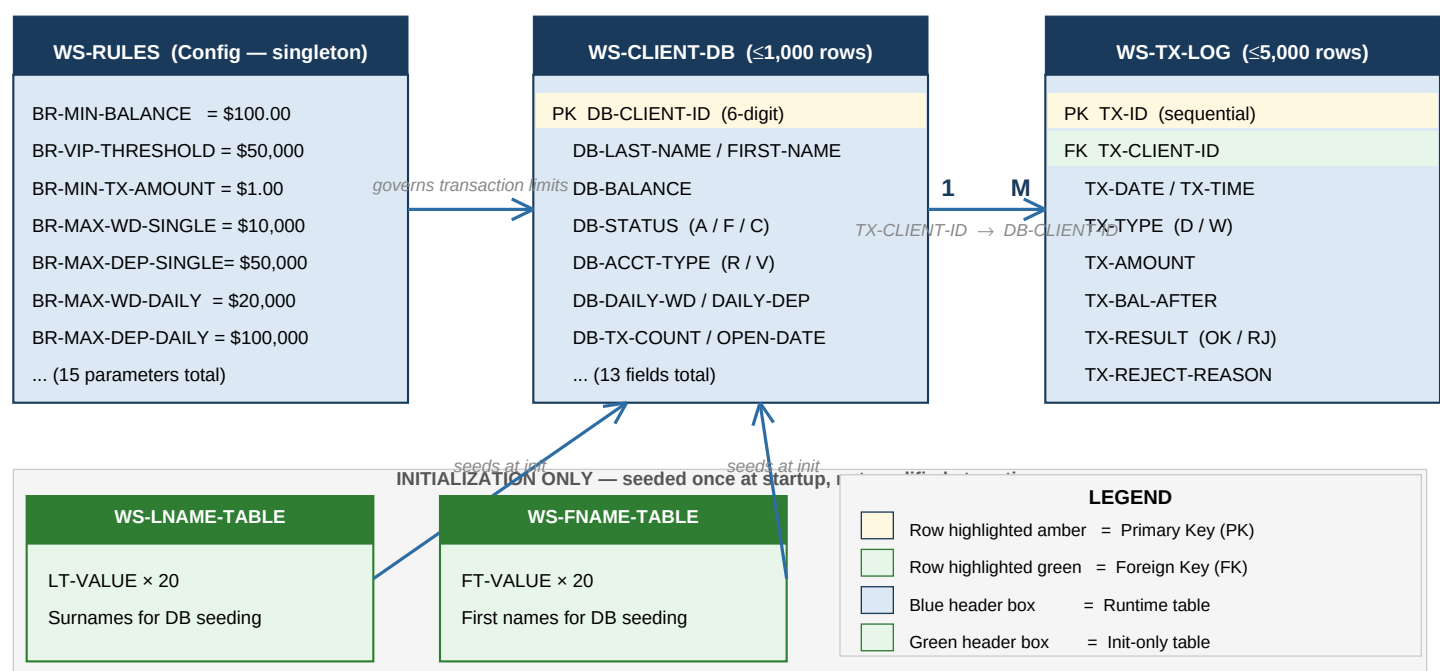
Rule Ref	Parameter	Current Value	COBOL Variable
BR-025	Minimum required balance	\$100.00	BR-MIN-BALANCE
BR-021	Regular max single withdrawal	\$10,000.00	BR-MAX-WD-SINGLE
BR-012	Regular max single deposit	\$50,000.00	BR-MAX-DEP-SINGLE
BR-023	Regular max daily withdrawal	\$20,000.00	BR-MAX-WD-DAILY
BR-014	Regular max daily deposit	\$100,000.00	BR-MAX-DEP-DAILY
BR-011/020	Minimum transaction amount	\$1.00	BR-MIN-TX-AMOUNT
BR-002	Initial account balance	\$500.00	BR-INITIAL-BALANCE
BR-008	VIP qualification threshold	\$50,000.00	BR-VIP-THRESHOLD
BR-022	VIP max single withdrawal	\$50,000.00	BR-VIP-MAX-WD-SINGLE
BR-013	VIP max single deposit	\$250,000.00	BR-VIP-MAX-DEP-SINGLE
BR-024	VIP max daily withdrawal	\$100,000.00	BR-VIP-MAX-WD-DAILY
BR-015	VIP max daily deposit	\$500,000.00	BR-VIP-MAX-DEP-DAILY

Key Design Decisions to Resolve During Migration

1. **VIP threshold logic is balance-based only.** There is no minimum tenure, credit score, or manual approval pathway — VIP status is granted and revoked automatically based solely on the balance crossing \$50,000.
2. **End-of-day reset is manual.** The current system requires a human operator to trigger the daily limit reset. The new system must decide whether this is automated (scheduled batch job) or remains operator-initiated.
3. **Rejected transactions are fully logged.** The audit trail includes both approved and rejected attempts. This behaviour must be preserved in the new system.
4. **No overdraft product exists.** The current system has no concept of an authorized overdraft. If the new system introduces overdraft, new business rules will be required.
5. **Frozen vs. Closed distinction.** Frozen is a temporary, reversible state that requires a branch visit to lift. Closed is permanent. The new system must preserve this distinction and implement the branch-visit workflow for unfreezing accounts.

14. Data Structures, Fields & Entity Relationships

The system holds three runtime in-memory databases and two initialization-only reference tables. The diagram below shows the structure and relationships; detailed field definitions follow.



14.1 WS-RULES — Business Rules Configuration

A single, read-only record loaded at startup. Contains every hardcoded threshold and limit applied during transaction processing. There is no key — the record is a singleton.

COBOL Field	Value	Data Type	Description
BR-MIN-BALANCE	\$100.00	Numeric 8.2	Minimum balance an account must retain after any withdrawal

COBOL Field	Value	Data Type	Description
BR-MAX-WD-SINGLE	\$10,000.00	Numeric 8.2	Maximum amount a Regular account may withdraw in one transaction
BR-MAX-DEP-SINGLE	\$50,000.00	Numeric 8.2	Maximum amount a Regular account may deposit in one transaction
BR-MAX-WD-DAILY	\$20,000.00	Numeric 8.2	Maximum total withdrawals a Regular account may make in one business day
BR-MAX-DEP-DAILY	\$100,000.00	Numeric 8.2	Maximum total deposits a Regular account may receive in one business day
BR-MIN-TX-AMOUNT	\$1.00	Numeric 8.2	Minimum dollar amount accepted for any transaction (deposit or withdrawal)
BR-INITIAL-BALANCE	\$500.00	Numeric 8.2	Opening balance assigned to every new account
BR-VIP-THRESHOLD	\$50,000.00	Numeric 8.2	Balance threshold above which an account is classified as VIP
BR-VIP-MAX-WD-SINGLE	\$50,000.00	Numeric 8.2	Maximum amount a VIP account may withdraw in one transaction
BR-VIP-MAX-DEP-SINGLE	\$250,000.00	Numeric 8.2	Maximum amount a VIP account may deposit in one transaction
BR-VIP-MAX-WD-DAILY	\$100,000.00	Numeric 8.2	Maximum total withdrawals a VIP account may make in one business day
BR-VIP-MAX-DEP-DAILY	\$500,000.00	Numeric 8.2	Maximum total deposits a VIP account may receive in one business day
BR-MAX-CLIENTS	1,000	Numeric 4	Maximum number of client accounts the database can hold
BR-MAX-TX-LOG	5,000	Numeric 5	Capacity of the transaction log circular buffer
BR-FIRST-CLIENT-ID	100001	Numeric 6	Starting value for auto-assigned Client IDs

14.2 WS-CLIENT-DB — Client Account Database

The primary account store. Holds up to 1,000 records, one per client. DB-CLIENT-ID is the primary key and the join field to WS-TX-LOG.

COBOL Field	Key	Data Type	Valid Values / Format	Description
DB-CLIENT-ID	PK	Numeric 6	100001 – 101000	Unique client identifier assigned sequentially at account opening
DB-LAST-NAME		Alpha 15	Free text	Client surname (up to 15 characters)
DB-FIRST-NAME		Alpha 10	Free text	Client given name (up to 10 characters)
DB-BALANCE		Signed Num 10.2	Any value $\geq$ BR-MIN-BALANCE	Current account balance in dollars
DB-STATUS		Alpha 1	A = Active F = Frozen C = Closed	Current account status — controls whether transactions are permitted
DB-ACCT-TYPE		Alpha 1	R = Regular V = VIP	Account tier — determines which transaction limits apply
DB-DAILY-WD		Numeric 10.2	0.00 – 999,999.99	Running total of withdrawals processed today; resets to 0 at end of day
DB-DAILY-DEP		Numeric 10.2	0.00 – 999,999.99	Running total of deposits received today; resets to 0 at end of day
DB-OPEN-DATE		Alpha 10	YYYY-MM-DD	Date the account was opened

COBOL Field	Key	Data Type	Valid Values / Format	Description
DB-TX-COUNT		Numeric 6	0 – 999999	Lifetime count of all transaction attempts (approved and rejected)
DB-LAST-TX-DATE		Alpha 10	YYYY-MM-DD	Date of the most recent transaction attempt on this account
DB-LAST-TX-TYPE		Alpha 1	D = Deposit W = Withdrawal	Type of the most recent transaction
DB-LAST-TX-AMT		Numeric 10.2	0.00 – 999,999.99	Dollar amount of the most recent transaction

14.3 WS-TX-LOG — Transaction Log

An audit log of every transaction attempt (approved and rejected). Implemented as a circular buffer of 5,000 entries — once full, the oldest entry is overwritten. TX-CLIENT-ID is a foreign key to WS-CLIENT-DB.DB-CLIENT-ID (many transactions per client).

COBOL Field	Key	Data Type	Valid Values / Format	Description
TX-ID	PK	Numeric 8	1, 2, 3 ... (never reset)	Sequential transaction identifier assigned at log time; never reused in a
TX-DATE		Alpha 10	YYYY-MM-DD	Date the transaction was processed
TX-TIME		Alpha 8	HH:MM:SS	Time the transaction was processed
TX-CLIENT-ID	FK	Numeric 6	100001 – 101000	Client account involved; joins to DB-CLIENT-ID in WS-CLIENT-DB
TX-TYPE		Alpha 1	D = Deposit W = Withdrawal	Type of transaction attempted
TX-AMOUNT		Numeric 10.2	0.00 – 999,999.99	Dollar amount of the transaction attempt
TX-BAL-AFTER		Signed Num 10.2	Any value	Account balance immediately after the transaction (populated for approv
TX-RESULT		Alpha 2	OK = Approved RJ = Rejected	Whether the transaction was approved or rejected by the validation rule
TX-REJECT-REASON		Alpha 30	Free text or spaces	Plain-text explanation of why the transaction was rejected; blank if approv

14.4 WS-LNAME-TABLE & WS-FNAME-TABLE — Initialization Seed Data

Two small lookup tables used only during system startup to populate the client database with sample names. They are not referenced during normal transaction processing and have no primary key or relationships at runtime.

Table	COBOL Field	Data Type	Entries	Purpose
WS-LNAME-TABLE	LT-VALUE	Alpha 15	20	Pool of surnames cycled across 1,000 client records at initialization
WS-FNAME-TABLE	FT-VALUE	Alpha 10	20	Pool of first names cycled across 1,000 client records at initialization

14.5 Entity Relationship Summary

From Table	Relationship	To Table	Join Condition	Cardinality
WS-RULES	governs	WS-CLIENT-DB	No join field — rules apply globally to all accounts	1 rule set → N accounts
WS-RULES	governs	WS-TX-LOG	No join field — rules govern every transaction	1 rule set → N log entries
WS-CLIENT-DB	referenced by	WS-TX-LOG	WS-TX-LOG.TX-CLIENT-ID = WS-CLIENT-DB.CLIENT-ID	1 account → N log entries
WS-LNAME-TABLE	seeds (init only)	WS-CLIENT-DB	MOD(record#, 20) selects surname index	20 names → 1,000 accounts
WS-FNAME-TABLE	seeds (init only)	WS-CLIENT-DB	MOD(record#+6, 20) selects first name index	20 names → 1,000 accounts